Check for updates

Quantifying Muscle Volume Deficits Among 38 Lower Extremity Muscles in Collegiate Football Athletes After Anterior Cruciate Ligament Reconstruction

Naoaki Ito, PT, DPT, PhD, Jack A. Martin, PhD, Mikel R. Joachim, PhD, Silvia S. Blemker, PhD, David A. Opar, PhD, Stephanie A. Kliethermes, PhD, Bryan C. Heiderscheit,* PT, PhD, and HAMIR Consortium Investigators
Investigation performed at the University of Wisconsin–Madison, Madison, Wisconsin, USA

Background: Quadriceps dysfunction is ubiquitous after anterior cruciate ligament (ACL) reconstruction (ACLR). Addressing quadriceps dysfunction is crucial to improve function, reduce the reinjury risk, and maintain long-term knee health. While deficits specific to the quadriceps are well documented, less is known about the effect of an ACL injury on other lower extremity muscle groups.

Purpose/Hypothesis: The purpose of this exploratory analysis was to quantify and rank lower extremity muscle volume deficits using magnetic resonance imaging in collegiate football athletes after ACLR. It was hypothesized that the quadriceps muscles would present with the greatest deficits and that compensatory hypertrophy of muscles at adjacent joints such as the hip and ankle would be observed.

Study Design: Cross-sectional study; Level of evidence, 3.

Methods: This study is a secondary analysis from an ongoing multicenter prospective cohort study involving Division I collegiate football athletes. Athletes who underwent primary unilateral ACLR (1 [3%] allograft, 2 [7%] quadriceps tendon autograft, 22 [73%] bone–patellar tendon–bone autograft, 5 [17%] hamstring tendon autograft) and magnetic resonance imaging were included. Muscle volumes ($mL \cdot kg^{-1} \cdot m^{-1}$) were quantified bilaterally from 38 lower extremity muscles using machine learning technology. Paired-samples *t* tests were performed between limbs for each muscle, which were then ranked and visualized in a forest plot based on standardized mean differences (surgical – nonsurgical limb).

Results: A total of 30 athletes (mean time from surgery, 27.9 ± 19.0 months) were included. The largest muscle volume deficits in the surgical limb were seen in the 3 uniarticular quadriceps muscles, followed by the biarticular triceps surae muscles. The rectus femoris and soleus did not show significant differences between limbs. Conversely, the fibularis muscle group had a greater muscle volume in the surgical limb compared with the nonsurgical limb. Most other muscle groups did not present significant differences between limbs.

Conclusion: Persistent quadriceps atrophy in a cohort of high-level athletes over 2 years after ACLR was highlighted in this study. Deficits in the gastrocnemius muscles, but not in the soleus, were also identified. This comprehensive approach examining various lower extremity muscles revealed latent muscle volume deficits present after ACLR.

Keywords: magnetic resonance imaging; machine learning; quadriceps; gastrocnemius; atrophy

After an anterior cruciate ligament (ACL) injury and ACL reconstruction (ACLR), athletes present with quadriceps weakness, inhibition, and atrophy.[19,50,52] The importance of regaining quadriceps strength has been repeatedly highlighted in the literature, as stronger quadriceps muscles lead to better function,[10,43,44] lower the reinjury risk,[13] and result in better long-term outcomes.[2,17] The current best practice for addressing quadriceps dysfunction includes the early introduction of open kinetic chain knee extension exercises[36] and neuromuscular electrical stimulation.[1] As rehabilitation progresses, complex multijoint movements such as jumps and hops that include coordination with the hip and ankle are also introduced. Although early rehabilitation often recommends strengthening the entire lower extremity,[3,5,33] isolated exercises targeting hip and ankle joints tend to receive less emphasis compared with isolated quadriceps strengthening.

Deficits in the quadriceps are well established,[15,20] but less is known about how other muscle groups are affected.[7] Graft type–specific concerns have led to some studies focusing on other muscle groups.[18,46,53] For example, athletes who have undergone ACLR using hamstring tendon

The American Journal of Sports Medicine
1–8
DOI: 10.1177/03635465241313235
© 2025 The Author(s)

autografts had significant atrophy and morphological changes in the semitendinosus muscle and tendon.[39,47,48,54] Apart from considerations for graft site morbidities, research has rarely considered changes to muscle morphology other than in the quadriceps. Athletes often develop compensatory movement patterns by shifting loads to the ipsilateral hip and ankle joints or to the contralateral limb to offset deficits at the knee after an ACL injury.[31,38,42,45] This is demonstrated via lower joint moments or excursions at the knee, while the trunk, hip, and ankle joints demonstrate greater moments or excursions during various tasks such as jumping, hopping, running, or walking.[6,11,24,28] It is therefore reasonable to hypothesize that, after ACLR, there may be a redistribution of muscle mass along the kinetic chain to reflect changes in the movement pattern.

Knowledge of the effect of an ACL injury and subsequent ACLR on specific muscle groups can benefit patients, as rehabilitation approaches can target muscle groups with deficits. This approach is often taken with the quadriceps,[5,33] but there may be other muscle groups that can benefit from similar isolated strengthening or neuromuscular electrical stimulation.[1] The primary limitation in investigating various muscle groups in the lower extremities has been the time-consuming nature of manual segmentation using imaging modalities such as magnetic resonance imaging (MRI)[26,51] or ultrasound[23,37] to quantify muscle morphology. With limited resources available, many researchers opt to focus their imaging-based research on the muscle group with the greatest known deficits, that is, the quadriceps, as knowledge gained from examining the quadriceps is expected to yield the greatest return on investment.

With advancements in technology, however, muscle volume quantification from various lower extremity muscles can now be achieved simultaneously, rapidly, accurately, and cost-effectively using machine learning algorithms.[14,35] Integrating such technology to examine patients after ACLR can enable a more rapid analysis of various lower extremity musculature, allowing researchers to investigate effects on a comprehensive set of muscles in a time-efficient manner. The purpose of this exploratory analysis was to quantify and rank lower extremity muscle volume deficits using MRI-based machine learning technology in collegiate football athletes after ACLR. We hypothesized that quadriceps muscles would have the greatest deficits and that compensatory hypertrophy of muscles at adjacent joints such as the hip and ankle would be observed.

METHODS

This study is a secondary analysis from an ongoing multicenter prospective cohort study involving Division I collegiate football athletes: "Hamstring Injury (HAMIR) Index: A Framework for Injury Mitigation Strategies Through Innovative Imaging, Biomechanics, and Data Analytics" (ClinicalTrials.gov: NCT05343052).[16] Athletes who underwent primary unilateral ACLR and lower extremity MRI as part of the parent study were included in this analysis. For athletes to be included, they must have been participating in football-related activities at full capacity at the time of MRI (ie, they were not injured or undergoing rehabilitation). Athletes with a history of multiple ACLR procedures were excluded. Surgical history and graft type for ACLR were self-reported by each athlete, and the medical team at each participating university confirmed the accuracy of details if necessary and if information was available. All participants enrolled in this study were informed of the risks and benefits of the parent study and signed an institutional review board–approved consent form.

MRI and Machine Learning–Driven Muscle Volume Calculation

MRI was conducted using a 3-T scanner (MAGNETOM Vida, Prisma, or Trio [Siemens] or SIGNA Premier [GE Healthcare]) equipped with a flexible surface coil. Imaging included lumbopelvic and bilateral lower extremity scans to measure muscle volumes throughout the entire field of view. Axial multislice 2-dimensional fast spoiled gradient recalled echo sequences were utilized, with parameters set to maximize the muscle signal and minimize the fat signal (repetition time/echo time: 839 ms/4.35 ms; field of view: 500 × 344 mm; matrix: 256 × 176; voxel size: 2 × 2 × 5 mm). Imaging was performed continuously from the 12th thoracic vertebra to below the ankle joint in sets of 40 scans. Automated muscle segmentation using Springbok Analytics' machine learning technology, which has demonstrated repeatability comparable with interobserver variability,[35] was employed to measure muscle volumes of all visible lower extremity muscles (see Appendix Figure A1, available in the online version of this article). The complete list of muscles segmented appears in the Results section. For standardized reporting, lower extremity muscle volumes were normalized to body weight and height ($mL \cdot kg^{-1} \cdot m^{-1}$).[14]

*Address correspondence to Bryan C. Heiderscheit, PT, PhD, Department of Orthopedics and Rehabilitation, University of Wisconsin–Madison, 1685 Highland Avenue, Madison, WI 53705, USA (email: heiderscheit@ortho.wisc.edu).

All authors are listed in the Authors section at the end of the article.

Submitted August 6, 2024; accepted November 20, 2024.

One or more of the authors has declared the following potential conflict of interest or source of funding: This study was supported by research funding from the National Football League (NFL). The NFL had no role in the study design, collection, analysis, or interpretation of the data presented and was not involved in the writing of the manuscript or in the decision to submit the manuscript for publication. S.S.B. (cofounder and Chief Science Officer) and B.C.H. (advisory board member) declare potential competing interests related to this work because of their roles with Springbok Analytics. AOSSM checks author disclosures against the Open Payments Database (OPD). AOSSM has not conducted an independent investigation on the OPD and disclaims any liability or responsibility relating thereto.

Statistical Analysis

Standard descriptive statistics for athlete age, athlete height, athlete weight, time from surgery in months, graft type, and athlete position were calculated and reported. No a priori power analysis was performed for this study, given the nature of a secondary analysis. Because the magnitude and distribution in volumes vary among all muscles, volumes were standardized to Z scores within each muscle for both limbs to compare across muscles. Paired-samples t tests were performed on Z scores between limbs (surgical vs nonsurgical) for the volumes of all 38 lower extremity muscles that were quantified. Muscles were then ranked based on standardized mean differences (surgical – nonsurgical) and displayed in a forest plot for visualization. Given the known influence of hamstring tendon autograft harvest on semitendinosus morphology,[34,47,54] a separate analysis excluding athletes who received hamstring tendon autografts was also performed. We opted for this approach over statistical modeling for the influence of graft type because of the small number of athletes who had undergone ACLR using a hamstring tendon autograft (n = 5).

RESULTS

A total of 30 male Division I collegiate football athletes met inclusion criteria and were included in this analysis (mean time from surgery, 27.9 ± 19.0 months; mean age, 21.0 ± 1.4 years; mean height, 1.85 ± 0.06 m; mean weight, 100.0 ± 15.3 kg) (1 [3%] allograft, 2 [7%] quadriceps tendon autograft, 22 [73%] bone–patellar tendon–bone autograft, 5 [17%] hamstring tendon autograft) (football position group[49]: 4 [13%] big, 13 [43%] power, 10 [33%] skill, 3 [10%] special teams).

Muscle Volumes Ranked by Standardized Mean Differences

The uniarticular quadriceps muscles (vastus intermedius, vastus lateralis, and vastus medialis) showed the greatest deficits in the surgical limb compared with the nonsurgical limb. The biarticular rectus femoris showed a similar deficit; however, the finding was not statistically significant. Deficits were also identified in the biarticular triceps surae muscles (medial and lateral gastrocnemii) but not in the uniarticular soleus. Conversely, the fibularis muscle group was the only muscle presenting a greater muscle volume in the surgical limb compared with the nonsurgical limb. The iliacus muscle also had deficits in the surgical limb, but only when athletes who received hamstring tendon autografts were removed. The remaining muscles presented no significant differences between limbs (Figure 1 and Appendix Table A1 [available online]).

While findings were not statistically significant, when all graft types were considered, the semitendinosus muscle appeared toward the bottom of the forest plot (mean difference, –0.15 $mL{\cdot}kg^{-1}{\cdot}m^{-1}$ [95% confidence interval (CI), –0.32 to 0.02 $mL{\cdot}kg^{-1}{\cdot}m^{-1}$]; P = .080), indicating a greater deficit in the surgical versus nonsurgical limb relative to other muscles (semitendinosus muscle volumes for the 5 athletes who received hamstring tendon autografts: surgical, 1.18 ± 0.60 $mL{\cdot}kg^{-1}{\cdot}m^{-1}$; nonsurgical, 2.10 ± 0.34 $mL{\cdot}kg^{-1}{\cdot}m^{-1}$). When athletes who received hamstring tendon autografts were removed, however, the semitendinosus muscle appeared toward the middle of the forest plot (mean difference, 0.00 $mL{\cdot}kg^{-1}{\cdot}m^{-1}$ [95% CI, –0.12 to 0.13 $mL{\cdot}kg^{-1}{\cdot}m^{-1}$]; P = .964), with no difference between the surgical and nonsurgical limbs (semitendinosus muscle volumes for the 25 athletes who did not receive hamstring tendon autografts: surgical, 2.02 ± 0.47 $mL{\cdot}kg^{-1}{\cdot}m^{-1}$; nonsurgical, 2.02 ± 0.47 $mL{\cdot}kg^{-1}{\cdot}m^{-1}$) (Figure 2 and Appendix Table A1).

DISCUSSION

The primary findings in this study reflect the ongoing struggle that athletes face in resolving quadriceps dysfunction after an ACL injury and subsequent ACLR. In a cohort of male Division I football athletes who had fully returned to sport after ACLR (mean time since surgery >2 years), the quadriceps muscles continued to have significant atrophy. Consistent with previous literature, the vastus muscles had greater deficits compared with the biarticular rectus femoris muscle.[4] Our knowledge on the importance of quadriceps strength,[13,17,27,29,40,41,43] combined with these findings, underscores the ongoing challenge in our field to best guide athletes back to their sport while minimizing the secondary injury risk and maximizing performance. It is crucial to continue to emphasize the importance of the early implementation of isolated quadriceps strengthening during rehabilitation, such as open kinetic chain knee extension exercises[36] and neuromuscular electrical stimulation,[1] which are known to improve quadriceps strength and promote hypertrophy. Furthermore, the present study findings reveal the necessity for continued quadriceps strengthening even after athletes return to sport. Quadriceps deficits persist for years, even among high-level athletes who have ample access to equipment, resources, and dedicated recovery time.

Gastrocnemius Muscle Volume Deficits

Surprisingly, muscle volume deficits were also observed in the ipsilateral gastrocnemius muscles. We initially hypothesized compensatory hypertrophy of the ipsilateral hip and ankle muscles due to athletes shifting loads away from the knee joint; however, the opposite effect was seen in the gastrocnemii. This may reflect overall unloading of the surgical limb, leading to disuse atrophy, even if a greater proportion of the loads within the surgical limb is shifted toward the ankle joint during various movements. The greater number of muscles placed on the left side of the forest plot (surgical limb deficit) may also reflect potential disuse atrophy in multiple muscle groups of the surgical limb.

Another interesting complementary finding in the ankle was the soleus muscle placing toward the top of the ranking (not statistically significant) (Figures 1 and 2). This indicated the potential opposite effect from the

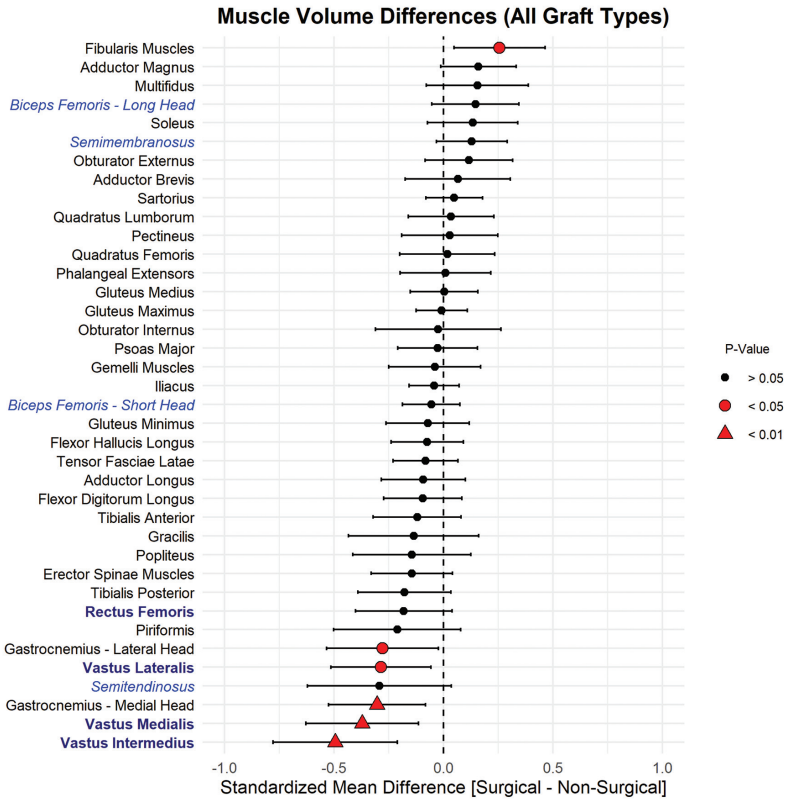

Figure 1. Paired *t* tests for muscle volume (*Z* score) differences between limbs ranked by standardized mean differences in athletes after anterior cruciate ligament reconstruction including all graft types (n = 30). Error bars indicate 95% confidence intervals. Quadriceps muscles are inboldface, and hamstring muscles are in italics.

gastrocnemius muscles, even when they both function as agonist muscles for plantarflexion with a common attachment via the Achilles tendon. When collectively considering ankle plantarflexion strength, a recent review demonstrated that no persistent weakness was found after ACLR.[12] In contrast, our findings may reflect each triceps surae muscle's unique contributions to knee arthrokinematics. The gastrocnemius muscles contribute to anterior translation of the tibia via the posterior attachment on the femur,[30] and the soleus muscle contributes to posterior translation of the tibia via the attachment on the proximal tibia.[8,21] While both muscles are considered primary plantarflexors of the ankle, only the gastrocnemius muscles that produce forces opposing resistance from the ACL had muscle volume deficits after ACLR. The findings from this study highlight muscle volume deficits from 2 major muscle groups known to contribute to anterior translation of the tibia: the quadriceps and the gastrocnemii.[30] These findings may be a result of greater posttraumatic muscle inhibition as a protective mechanism in the knee joint, which lacks the native passive resistance against tibial anterior translation.

Fibularis Muscle Hypertrophy

Only the combined fibularis muscles had a statistically significantly greater muscle volume in the surgical limb compared with the nonsurgical limb. To our knowledge, no literature exists to date on this muscle group after an ACL injury. The fibularis muscles are primary evertors of the ankle with a synergistic role in ankle plantarflexion. A potential explanation for compensatory hypertrophy in this muscle group may be increased demands in frontal-plane control during unilateral movements. Because of known quadriceps dysfunction and stiff knee movement patterns (ie, limited knee flexion during movements),[22,25,31] the center of mass is higher during unilateral movements in athletes after ACLR. This may lead to a longer moment arm for ankle frontal-plane control, resulting in higher demands of the fibularis muscle group. Alternatively, along with the soleus, the fibularis muscles could be compensating for plantarflexion deficits due to gastrocnemius atrophy. The clinical implications are unknown, and the benefit of further exploring this finding is uncertain; however, this finding may provide insight for future research.

Minimal Muscle Volume Deficits in Many Muscles

When interpreting findings from exploratory analyses such as the one performed in this study, it is easy to focus on the "statistically significant" effects and ignore the null findings. We must however appreciate that, apart from the expected findings (ie, quadriceps and semitendinosus) and

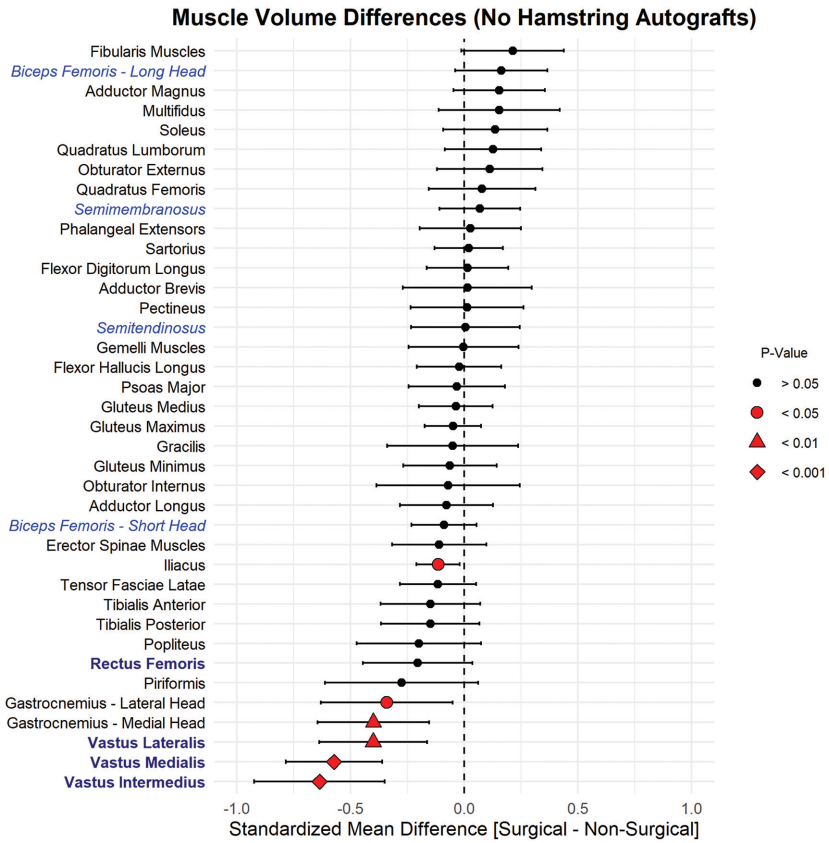

Figure 2. Paired *t* tests for muscle volume (*Z* score) differences between limbs ranked by standardized mean differences in athletes after anterior cruciate ligament reconstruction excluding hamstring tendon autografts (n = 25). Error bars indicate 95% confidence intervals. Quadriceps muscles are in boldface, and hamstring muscles in italics.

a few exceptions discussed above, most muscle groups tended not to have any muscle volume deficits once high-level athletes had returned to sport. This includes no notable hip muscle volume differences between limbs, even while persistent biomechanical compensations were commonly observed at the hip joint.[31] This finding is consistent with research on hip strength after ACLR, which identified no meaningful hip weakness after ACLR.[12] Because of the cross-sectional nature of this analysis, it is unknown whether such findings resulted from the muscle groups not being influenced by an ACL injury or the success of current rehabilitation standards. Nonetheless, the null findings in most muscles allow future research and potential rehabilitation protocols to further emphasize a narrower set of muscle groups.

Novel Approach Revealing Latent Muscle Groups With Deficits

By considering all lower extremity muscles in athletes after ACLR, we revealed deficits in secondarily affected muscles that may not have been captured using traditional methodology, with the potential to be addressed by targeted interventions. Beyond the injury-specific findings of this study, our comprehensive approach examining a large set of muscles can serve as a model for efficiently and effectively expanding our knowledge of lower extremity muscle deficits after various injuries. In other common lower extremity injuries such as ankle sprains[32] or femoroacetabular impingement,[9] most evidence and treatment have also traditionally focused on the muscles at the involved joint. The approach applied in this cohort of athletes after ACLR can provide a more comprehensive way to consider the entire lower extremities. This broader perspective can lead to clinical insights that may have been largely overlooked because of previous methodological limitations.

Specific to this study, possible atrophy of ankle musculature uncovered the potential need for rehabilitation protocols after ACLR to place greater emphasis on plantarflexion strengthening. While quadriceps strengthening is still the priority, the addition of isolated ankle plantarflexion strengthening can be an excellent adjunct to address the deficits that we observed. Many rehabilitation protocols today incorporate complex multijoint movements such as squats, lunges, or various types of jumps to strengthen the lower extremity kinetic chain.[5,33] However, not many protocols specifically recommend isolated ankle strengthening using exercises such as heel raises with heavy resistance. Similar to the favorable effects that we have observed via the early implementation of

heavy open kinetic chain knee extension exercises in rehabilitation,[36] there may also be benefits to adding isolated plantarflexion strengthening to our current standard of care after ACLR. While more work is needed to understand the specific implications of gastrocnemius muscle volume deficits after ACLR, we encourage clinicians and researchers to begin exploring the implementation of isolated plantarflexion strengthening exercises with athletes after an ACL injury.

Limitations

There are limitations to consider in this study. As an ongoing prospective cohort study focusing on a specific population, this sample is limited to young high-level male football athletes. While the homogeneity of the cohort may be a strength when working with high-level male football athletes, readers should be cautious about applying the findings from this study to other populations. Our study is also limited to investigating muscle volumes, which are strongly correlated with strength[26,51] but are not a direct measure of strength or function. As seen in our findings from the triceps surae muscle group, however, it may be important to continue examining both joint- and muscle-level deficits because the specific muscle influenced may not always be reflected in joint-level performance. Further considerations for graft type and time from surgery are also necessary. The known influence of hamstring tendon autograft harvest[47,54] was clearly reflected even in our limited sample, but confirmation with a larger cohort is needed. The athletes in this study were all participating at a high level at the time of enrollment; however, there was still a notable range in time from surgery, which influences recovery. In our cross-sectional cohort, the limb symmetry index ([surgical/nonsurgical] $\times$ 100) of the quadriceps muscle volumes (vastus medialis: $R^2 = 0.489$, $P < .01$; vastus lateralis: $R^2 = 0.218$, $P < .01$; vastus intermedius: $R^2 = 0.193$, $P < .01$; rectus femoris: $R^2 = 0.016$, $P = .48$) had a positive linear relationship with time from surgery, while that of the gastrocnemius muscle volumes (medial gastrocnemius: $R^2 = 0.035$, $P = .29$; lateral gastrocnemius: $R^2 = 0.001$, $P = .82$) did not. A study intended to capture longitudinal changes in muscle volumes should also be performed in the future to further explore these findings. We also acknowledge the relatively small sample size in comparison to the number of analyses that were performed. Type I errors may have contributed to our findings in the fibularis muscles, and the lack of findings in other muscles, particularly the ones presenting with relatively large effect sizes but with wide 95% CIs, may be a result of type II errors.

CONCLUSION

Persistent quadriceps atrophy was highlighted after ACLR in a cohort of high-level male football athletes who had returned to sport at full capacity. Continued attention to quadriceps strengthening must be emphasized during rehabilitation and after returning to play for improvements in performance, the prevention of secondary injuries, and the maintenance of long-term knee joint health. Gastrocnemius muscle deficits were also found in this exploratory analysis, indicating the potential need to emphasize plantarflexion strengthening after an ACL injury. Clinical implications of findings at the ankle, however, must be further investigated in relation to performance or outcome measures before definitive recommendations can be made. Including targeted gastrocnemius muscle strengthening during rehabilitation is extremely simple and carries a minimal risk; hence, clinicians and future studies should consider incorporating isolated ankle muscle strengthening while continuing to place the primary focus on the quadriceps.

AUTHORS

Naoaki Ito, PT, DPT, PhD (Department of Orthopedics and Rehabilitation, University of Wisconsin–Madison, Madison, Wisconsin, USA; Badger Athletic Performance Program, University of Wisconsin–Madison, Madison, Wisconsin, USA); Jack A. Martin, PhD (Department of Orthopedics and Rehabilitation, University of Wisconsin–Madison, Madison, Wisconsin, USA; Badger Athletic Performance Program, University of Wisconsin–Madison, Madison, Wisconsin, USA; Department of Mechanical Engineering, University of Wisconsin–Madison, Madison, Wisconsin, USA); Mikel R. Joachim, PhD (Department of Orthopedics and Rehabilitation, University of Wisconsin–Madison, Madison, Wisconsin, USA; Badger Athletic Performance Program, University of Wisconsin–Madison, Madison, Wisconsin, USA); Silvia S. Blemker, PhD (Springbok Analytics, Charlottesville, Virginia, USA; Department of Biomedical Engineering, University of Virginia, Charlottesville, Virginia, USA); David A. Opar, PhD (Sports Performance, Recovery, Injury and New Technologies Research Centre, Australian Catholic University, Melbourne, Victoria, Australia; School of Behavioural and Health Sciences, Australian Catholic University, Melbourne, Victoria, Australia); Stephanie A. Kliethermes, PhD (Department of Orthopedics and Rehabilitation, University of Wisconsin–Madison, Madison, Wisconsin, USA; Badger Athletic Performance Program, University of Wisconsin–Madison, Madison, Wisconsin, USA); Bryan C. Heiderscheit, PT, PhD (Department of Orthopedics and Rehabilitation, University of Wisconsin–Madison, Madison, Wisconsin, USA; Badger Athletic Performance Program, University of Wisconsin–Madison, Madison, Wisconsin, USA).

HAMIR Consortium Investigators:

Geoffrey Baer, PhD, MD, (Department of Orthopedics and Rehabilitation, University of Wisconsin–Madison, Madison, WI, USA; Badger Athletic Performance Program, University of Wisconsin–Madison, Madison, WI, USA), Kurrel Fabian, DAT, (Badger Athletic Performance Program, University of Wisconsin–Madison, Madison, WI, USA), Erin Hammer, MD, MPH, (Department of Orthopedics and Rehabilitation, University of Wisconsin–Madison, Madison, WI, USA; Badger Athletic Performance Program, University of Wisconsin–Madison, Madison, WI, USA), Danielle Heidt, MS, LAT, (Badger Athletic Performance Program, University of Wisconsin–Madison, Madison, WI, USA), Matthew Kuehl, MS, LAT, (Badger Athletic Performance Program, University of Wisconsin–Madison, Madison, WI, USA), Kenneth Lee, MD, MBA, (Department of Radiology, University of Wisconsin–Madison, Madison, WI, USA), Michael Moll, MEd, LAT, (Badger Athletic Performance Program,

University of Wisconsin–Madison, Madison, WI, USA), Ryan Peot, MS, LAT, (Badger Athletic Performance Program, University of Wisconsin–Madison, Madison, WI, USA), Jennifer Sanfilippo, PhD, LAT, (Badger Athletic Performance Program, University of Wisconsin–Madison, Madison, WI, USA), Claire Tanaka, MS, (Department of Orthopedics and Rehabilitation, University of Wisconsin–Madison, Madison, WI, USA; Badger Athletic Performance Program, University of Wisconsin–Madison, Madison, WI, USA), Sarah Sund, MS, (Department of Orthopedics and Rehabilitation, University of Wisconsin–Madison, Madison, WI, USA), Yi-Chung Lin, PhD, (Sports Performance, Recovery, Injury and New Technologies (SPRINT) Research Centre, Australian Catholic University, Melbourne, Australia; School of Behavioral and Health Sciences, Australian Catholic University, Melbourne, Australia), Jack Hickey, PhD, (Sports Performance, Recovery, Injury and New Technologies (SPRINT) Research Centre, Australian Catholic University, Melbourne, Australia; School of Behavioral and Health Sciences, Australian Catholic University, Melbourne, Australia), Ashleigh Homer, PhD, (Sports Performance, Recovery, Injury and New Technologies (SPRINT) Research Centre, Australian Catholic University, Melbourne, Australia; School of Behavioral and Health Sciences, Australian Catholic University, Melbourne, Australia), Nirav Maniar, PhD, (Sports Performance, Recovery, Injury and New Technologies (SPRINT) Research Centre, Australian Catholic University, Melbourne, Australia; School of Behavioral and Health Sciences, Australian Catholic University, Melbourne, Australia), Ryan Timmins, PhD, (Sports Performance, Recovery, Injury and New Technologies (SPRINT) Research Centre, Australian Catholic University, Melbourne, Australia; School of Behavioral and Health Sciences, Australian Catholic University, Melbourne, Australia), Matthew Cousins, BS, (Springbok Analytics, USA), Olivia DuCharme, ME, (Springbok Analytics, USA), Xue Feng, PhD, (Springbok Analytics, USA), Anthony Nguyen, BS, (Springbok Analytics, USA), Lara Riem, PhD, (Springbok Analytics, USA), Joseph Hart, PhD, ATC, (University of North Carolina, Chapel Hill, NC, USA), Noah McCoy, MS, University of North Carolina, Chapel Hill, NC, USA), Dain Allred, MD, (Brigham Young University, Provo, UT, USA), Anthony Beutler, MD, (Brigham Young University, Provo, UT, USA), Darren Campbell, MD, (Brigham Young University, Provo, UT, USA), A. Wayne Johnson, PT, PhD, (Brigham Young University, Provo, UT, USA), Malorie Wilwand, MS, (Brigham Young University, Provo, UT, USA), Brett Mortensen, PT, PhD, ATC, (Brigham Young University, Provo, UT, USA), Emma Remington, LAT, (Brigham Young University, Provo, UT, USA), Jonathon Hauenstein, PhD, (University of Notre Dame, South Bend, IN, USA), Casey Metoyer, PhD, (University of Notre Dame, South Bend, IN, USA), John Wagle, PhD, (University of Notre Dame, South Bend, IN, USA), Josh Huff, BS, (Indiana University, Bloomington, IN, USA), Nicholas Port, PhD, (Indiana University, Bloomington, IN, USA), Jordan Reyes, MS (Indiana University, Bloomington, IN, USA).

ACKNOWLEDGMENT

This study was made possible through the efforts of those affiliated with the HAMIR project.

ORCID iDs

Naoaki Ito https://orcid.org/0000-0003-4166-5319
Mikel R. Joachim https://orcid.org/0000-0002-3085-4491
David A. Opar https://orcid.org/0000-0002-8354-6353

REFERENCES

1. Arhos EK, Ito N, Hunter-Giordano A, Nolan TP, Snyder-Mackler L, Silbernagel KG. Who's afraid of electrical stimulation? Let's revisit the application of NMES at the knee. *J Orthop Sports Phys Ther*. 2024;54(2):1-6.
2. Arhos EK, Thoma LM, Grindem H, Logerstedt D, Risberg MA, Snyder-Mackler L. Association of quadriceps strength symmetry and surgical status with clinical osteoarthritis five years after anterior cruciate ligament rupture. *Arthritis Care Res (Hoboken)*. 2022;74(3):386-391.
3. Arundale AJH, Bizzini M, Dix C, et al. Exercise-based knee and anterior cruciate ligament injury prevention. *J Orthop Sports Phys Ther*. 2023;53(1):CPG1-CPG34.
4. Birchmeier T, Lisee C, Kane K, Brazier B, Triplett A, Kuenze C. Quadriceps muscle size following ACL injury and reconstruction: a systematic review. *J Orthop Res*. 2020;38(3):598-608.
5. Brinlee AW, Dickenson SB, Hunter-Giordano A, Snyder-Mackler L. ACL reconstruction rehabilitation: clinical data, biologic bealing, and criterion-based milestones to inform a return-to-sport guideline. *Sports Health*. 2022;14(5):770-779.
6. Capin JJ, Zarzycki R, Ito N, et al. Gait mechanics in women of the ACL-SPORTS randomized control trial: interlimb symmetry improves over time regardless of treatment group. *J Orthop Res*. 2019;37(8):1743-1753.
7. Dutaillis B, Maniar N, Opar DA, Hickey JT, Timmins RG. Lower limb muscle size after anterior cruciate ligament injury: a systematic review and meta-analysis. *Sports Med*. 2021;51(6):1209-1226.
8. Elias JJ, Faust AF, Chu YH, Chao EY, Cosgarea AJ. The soleus muscle acts as an agonist for the anterior cruciate ligament: an in vitro experimental study. *Am J Sports Med*. 2003;31(2):241-246.
9. Enseki KR, Bloom NJ, Harris-Hayes M, et al. Hip pain and movement dysfunction associated with nonarthritic hip joint pain: a revision. *J Orthop Sports Phys Ther*. 2023;53(7):CPG1-CPG70.
10. Garcia SA, Moffit TJ, Vakula MN, Holmes SC, Montgomery MM, Pamukoff DN. Quadriceps muscle size, quality, and strength and self-reported function in individuals with anterior cruciate ligament reconstruction. *J Athl Train*. 2020;55(3):246-254.
11. Gill VS, Tummala SV, Han W, et al. Athletes continue to show functional performance deficits at return to sport after anterior cruciate ligament reconstruction: a systematic review. *Arthroscopy*. 2024;40(8):2309-2321.e2.
12. Girdwood M, Culvenor AG, Patterson B, et al. No sign of weakness: a systematic review and meta-analysis of hip and calf muscle strength after anterior cruciate ligament injury. *Br J Sports Med*. 2024;58(9):500-510.
13. Grindem H, Snyder-Mackler L, Moksnes H, Engebretsen L, Risberg MA. Simple decision rules can reduce reinjury risk by 84% after ACL reconstruction: the Delaware-Oslo ACL cohort study. *Br J Sports Med*. 2016;50(13):804-808.
14. Handsfield GG, Meyer CH, Hart JM, Abel MF, Blemker SS. Relationships of 35 lower limb muscles to height and body mass quantified using MRI. *J Biomech*. 2014;47(3):631-638.
15. Hart JM, Pietrosimone B, Hertel J, Ingersoll CD. Quadriceps activation following knee injuries: a systematic review. *J Athl Train*. 2010;45(1):87-97.
16. Heiderscheit BC, Blemker SS, Opar D, et al. The development of a HAMstring InjuRy (HAMIR) index to mitigate injury risk through innovative imaging, biomechanics, and data analytics: protocol for an observational cohort study. *BMC Sports Sci Med Rehabil*. 2022;14(1):128.
17. Hipsley A, Hall M, Saxby DJ, Bennell KL, Wang X, Bryant AL. Quadriceps muscle strength at 2 years following anterior cruciate ligament reconstruction is associated with tibiofemoral joint cartilage volume. *Knee Surg Sports Traumatol Arthrosc*. 2022;30(6):1949-1957.
18. Huber R, Viecelli C, Bizzini M, et al. Knee extensor and flexor strength before and after anterior cruciate ligament reconstruction in a large sample of patients: influence of graft type. *Phys Sportsmed*. 2019;47(1):85-90.
19. Hughes JD, Burnham JM, Hirsh A, et al. Comparison of short-term Biodex results after anatomic anterior cruciate ligament

reconstruction among 3 autografts. *Orthop J Sports Med*. 2019;7(5):2325967119847630.
20. Hunnicutt JL, McLeod MM, Slone HS, Gregory CM. Quadriceps neuromuscular and physical function after anterior cruciate ligament reconstruction. *J Athl Train*. 2020;55(3):238-245.
21. Hurd WJ, Snyder-Mackler L. Knee instability after acute ACL rupture affects movement patterns during the mid-stance phase of gait. *J Orthop Res*. 2007;25(10):1369-1377.
22. Ito N, Capin JJ, Khandha A, Buchanan TS, Snyder-Mackler L. Identifying gait pathology after ACL reconstruction using temporal characteristics of kinetics and electromyography. *Med Sci Sports Exerc*. 2022;54(6):923-930.
23. Ito N, Sigurðsson HB, Snyder-Mackler L, Grävare Silbernagel K. Time to treat the tendon rupture induced by surgery: early hypertrophy of the patellar tendon graft site predicts strong quadriceps after ACLR with bone-patellar tendon-bone autograft. *Knee Surg Sports Traumatol Arthrosc*. 2023;31(12):5791-5798.
24. Johnston PT, McClelland JA, Webster KE. Lower limb biomechanics during single-leg landings following anterior cruciate ligament reconstruction: a systematic review and meta-analysis. *Sports Med*. 2018;48(9):2103-2126.
25. Kaur M, Ribeiro DC, Theis JCC, Webster KE, Sole G. Movement patterns of the knee during gait following ACL reconstruction: a systematic review and meta-analysis. *Sports Med*. 2016;46(12):1869-1895.
26. Kuenze CM, Blemker SS, Hart JM. Quadriceps function relates to muscle size following ACL reconstruction. *J Orthop Res*. 2016;34(9):1656-1662.
27. Lewek M, Rudolph K, Axe M, Snyder-Mackler L. The effect of insufficient quadriceps strength on gait after anterior cruciate ligament reconstruction. *Clin Biomech*. 2002;17(1):56-63.
28. Lindström M, Felländer-Tsai L, Wredmark T, Henriksson M. Adaptations of gait and muscle activation in chronic ACL deficiency. *Knee Surg Sports Traumatol Arthrosc*. 2010;18(1):106-114.
29. Logerstedt D, Lynch A, Axe MJ, Snyder-Mackler L. Pre-operative quadriceps strength predicts IKDC2000 scores 6 months after anterior cruciate ligament reconstruction. *Knee*. 2013;20(3):208-212.
30. Maniar N, Cole MH, Bryant AL, Opar DA. Muscle force contributions to anterior cruciate ligament loading. *Sports Med*. 2022;52(8):1737-1750.
31. Markström JL, Tengman E, Hager CK. ACL-reconstructed and ACL-deficient individuals show differentiated trunk, hip, and knee kinematics during vertical hops more than 20 years post-injury. *Knee Surg Sports Traumatol Arthrosc*. 2018;26(2):358-367.
32. Martin RRL, Davenport TE, Fraser JJ, et al. Ankle stability and movement coordination impairments: lateral ankle ligament sprains revision 2021. *J Orthop Sports Phys Ther*. 2021;51(4):CPG1-CPG80.
33. Mayer MA, Deliso M, Hong IS, et al. Rehabilitation and return to play protocols after anterior cruciate ligament reconstruction in soccer players: a systematic review. *Am J Sports Med*. 2025;53(1):217-227.
34. Messer DJ, Shield AJ, Williams MD, Timmins RG, Bourne MN. Hamstring muscle activation and morphology are significantly altered 1–6 years after anterior cruciate ligament reconstruction with semitendinosus graft. *Knee Surg Sports Traumatol Arthrosc*. 2020;28(3):733-741.
35. Ni R, Meyer CH, Blemker SS, Hart JM, Feng X. Automatic segmentation of all lower limb muscles from high-resolution magnetic resonance imaging using a cascaded three-dimensional deep convolutional neural network. *J Med Imaging*. 2019;6(4):044009.
36. Noehren B, Snyder-Mackler L. Who's afraid of the big bad wolf? Open-chain exercises after anterior cruciate ligament reconstruction. *J Orthop Sports Phys Ther*. 2020;50(9):473-475.
37. Noorkoiv M, Nosaka K, Blazevich AJ. Assessment of quadriceps muscle cross-sectional area by ultrasound extended-field-of-view imaging. *Eur J Appl Physiol*. 2010;109(4):631-639.
38. Pairot De Fontenay B, Argaud S, Blache Y, Monteil K. Asymmetries in joint work during multi-joint movement after anterior cruciate ligament reconstruction: a pilot study. *Scand J Med Sci Sports*. 2000;24(6):471-476.
39. Papandrea P, Vulpiani MC, Ferretti A, Conteduca F. Regeneration of the semitendinosus tendon harvested for anterior cruciate ligament reconstruction: evaluation using ultrasonography. *Am J Sports Med*. 2000;28(4):556-561.
40. Pietrosimone B, Davis-Wilson HC, Seeley MK, et al. Gait biomechanics in individuals meeting sufficient quadriceps strength cutoffs following anterior cruciate ligament reconstruction. *J Athl Train*. 2021;56(9):960-966.
41. Qiu J, Ong MTY, Choi CY, et al. Associations of patient characteristics, rate of torque development, voluntary activation of quadriceps with quadriceps strength, and knee function before anterior cruciate ligament reconstruction. *Res Sports Med*. 2024;32(3):491-503.
42. Roos PE, Button K, van Deursen RW. Motor control strategies during double leg squat following anterior cruciate ligament rupture and reconstruction: an observational study. *J Neuroeng Rehabil*. 2014;11:19.
43. Schmitt LC, Paterno MV, Hewett TE. The impact of quadriceps femoris strength asymmetry on functional performance at return to sport following anterior cruciate ligament reconstruction. *J Orthop Sports Phys Ther*. 2012;42(9):750-759.
44. Shi H, Huang H, Ren S, et al. The relationship between quadriceps strength asymmetry and knee biomechanics asymmetry during walking in individuals with anterior cruciate ligament reconstruction. *Gait Posture*. 2019;73(May):74-79.
45. Sigward SM, Chan MM, Lin PE, Almansouri SY, Pratt KA. Compensatory strategies that reduce knee extensor demand during a bilateral squat change from 3 to 5 months following anterior cruciate ligament reconstruction. *J Orthop Sports Phys Ther*. 2018;48(9):713-718.
46. Smith AH, Capin J, Zarzycki R, Snyder-Mackler L. Athletes with bone-patellar tendon-bone autograft for ACL reconstruction were months slower to meet rehabilitation milestones and return to sport criteria than athletes with hamstring tendon autograft or soft tissue allograft: secondary analysis from the ACL-SPORTS trial. *J Orthop Sports Phys Ther*. 2020;50(5):259-266.
47. Speedtsberg MB, Zebis MK, Lauridsen HB, Magnussen E, Hölmich P. Anatomical retraction of the semitendinosus muscle following harvest of the distal semitendinosus tendon for ACL reconstruction. *Knee Surg Sports Traumatol Arthrosc*. 2022;30(5):1706-1710.
48. Suydam SM, Cortes DH, Axe MJ, Snyder-Mackler L, Buchanan TS. Semitendinosus tendon for ACL reconstruction: regrowth and mechanical property recovery. *Orthop J Sports Med*. 2017;5(6):2325967117712944.
49. Tanaka CS, Stiffler-Joachim MR, Merrill EA, et al. Eccentric hamstring strength by position for Division 1 football athletes. *Med Sci Sports Exerc*. 2023;55(9S):715-716.
50. Tayfur B, Charuphongsa C, Morrissey D, Miller SC. Neuromuscular function of the knee joint following knee injuries: does it ever get back to normal? A systematic review with meta-analyses. *Sports Med*. 2020;51(2):321-338.
51. Thomas AC, Wojtys EM, Brandon C, Palmieri-Smith RM. Muscle atrophy contributes to quadriceps weakness after ACL reconstruction. *J Sci Med Sport*. 2016;19(1):7-11.
52. Tsai LC, Jeanfreau CM, Hamblin KA, et al. Time, graft, sex, geographic location, and isokinetic speed influence the degree of quadriceps weakness after anterior cruciate ligament reconstruction: a systematic review and meta-analysis. *Knee Surg Sports Traumatol Arthrosc*. 2022;30(10):3367-3376.
53. Widner M, Dunleavy M, Lynch S. Outcomes following ACL reconstruction based on graft type: are all grafts equivalent? *Curr Rev Musculoskelet Med*. 2019;12(4):460-465.
54. Williams GN, Snyder-Mackler L, Barrance PJ, Axe MJ, Buchanan TS. Muscle and tendon morphology after reconstruction of the anterior cruciate ligament with autologous semitendinosus-gracilis graft. *J Bone Joint Surg Am*. 2004;86(9):1936-1946.

For reprints and permission queries, please visit Sage's Web site at http://www.sagepub.com/journals-permissions